group 1

Understanding Inflation in India

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STATISTICAL OVERVIEW

- Inflation Trends in India (2014-2024)
  - Influenced by geopolitical, economic, and environmental factors.
  - Key events: crude oil price crash, demonetization, COVID-19 pandemic.
  - Impact of RBI policies, weather conditions, and global supply chain disruptions.
- Global Economic Challenges & India’s Resilience
  - Uncertainties: pandemic, trade tensions, supply chain shifts.
  - Monetary tightening in the US & EU; economic slowdown in major economies.
  - India’s economic recovery: rising IIP, GST collections, and consumer confidence.
  - IMF projects India as a key driver of global growth.
- State-Level Inflation Analysis (2014-2016)
  - Focus on Maharashtra, Uttar Pradesh, Tamil Nadu, and West Bengal.
  - Major inflation drivers: housing costs, fuel prices, and miscellaneous expenses.
  - Relationship between food inflation, rainfall patterns, and trade dynamics.
- Policy Implications & Economic Stability
  - Understanding inflation dynamics for targeted policy measures.
  - Need for a nuanced approach considering regional economic conditions.
  - Aim: Effective policies to manage inflation and ensure economic stability.

About Dataset

To analyze inflation trends in India, we have combined data from three key sources:

1. Consumer Price Index (CPI) – Tracking Inflation
   - Shows price changes in essential goods like food, housing, and fuel across rural and urban areas.
   - Helps us understand inflation trends across states.

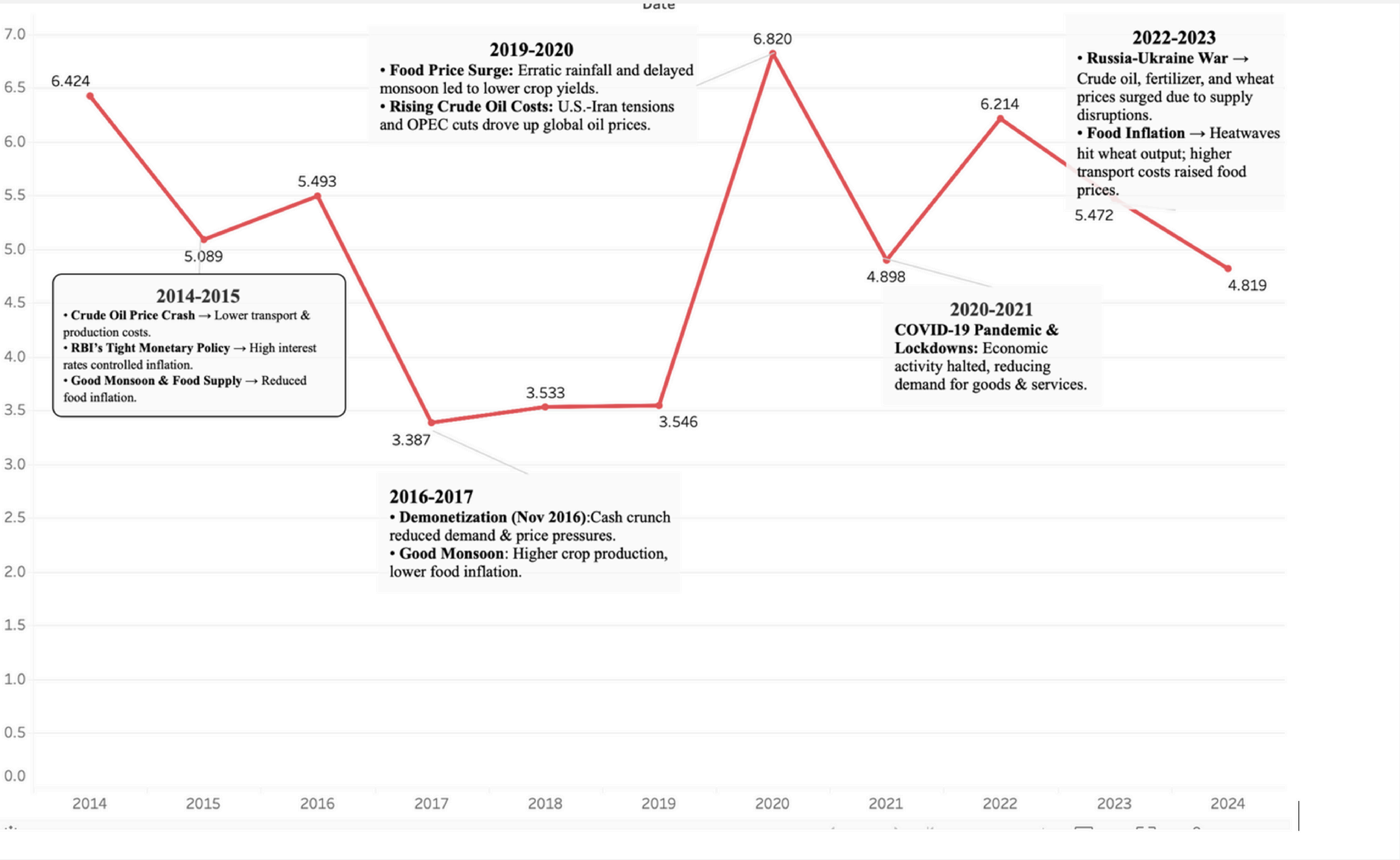
   Source: Ministry of Statistics and Program Implementation (MOSPI).
2. Rainfall Data – Impact on Agriculture
   - Tracks annual rainfall in 36 regions of India.
   - Helps analyze how weather affects agriculture and food prices.

   Source: Indian Meteorological Department (IMD) & RBI.
3. Trade Data – Imports & Exports
   - Covers India's imports and exports (2014-2024) by commodity.
   - Shows how global trade impacts domestic inflation.

   Source: Ministry of Commerce and Industry.

Theme 1: Inflation Trends in India (2014-2024): Key Drivers and Events

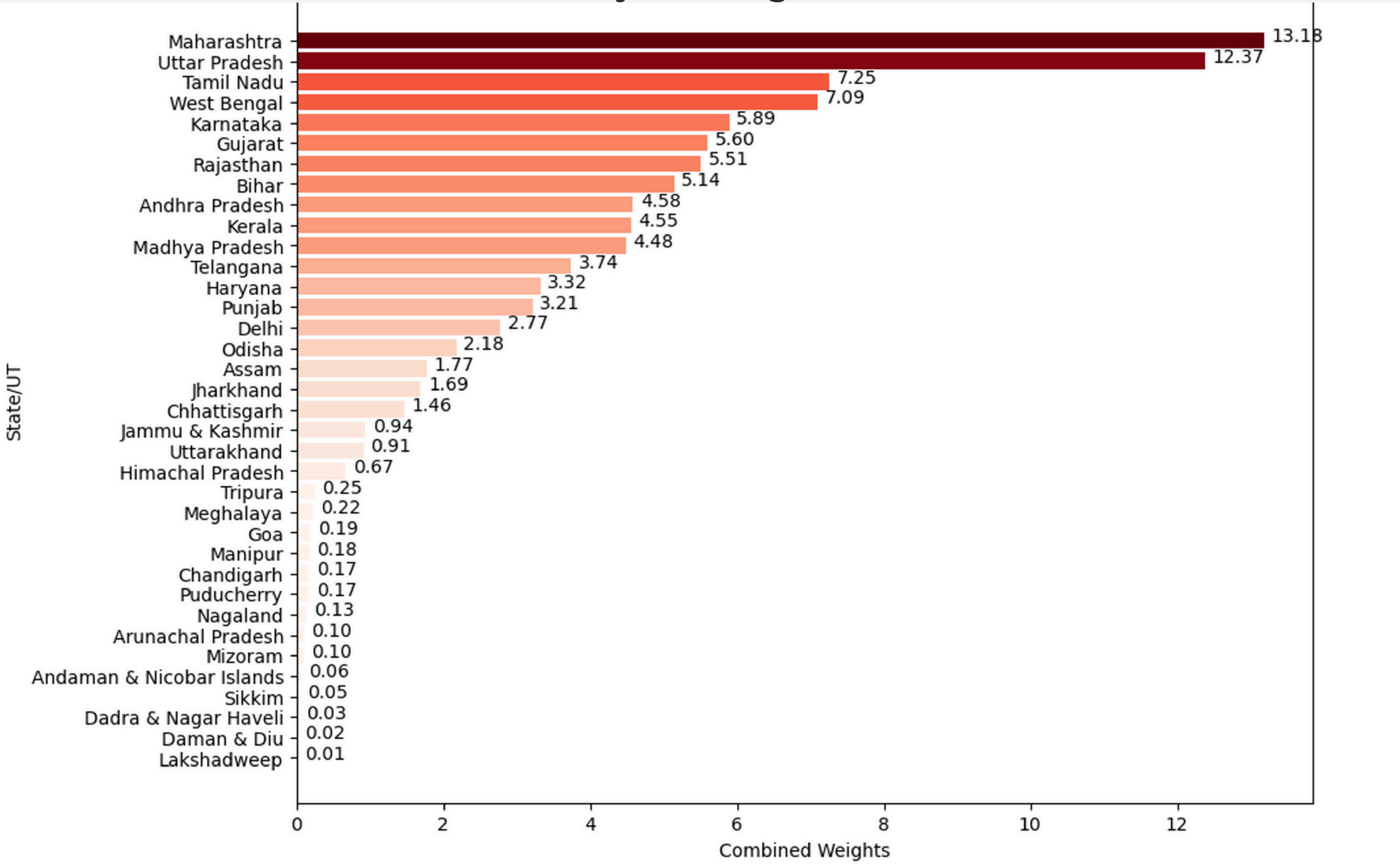
Chart 1: Inflation Trends Across Indian States



It illustrates the average inflation rate in India from 2014 to 2024, highlighting the key events and factors that influenced these trends. The data shows significant fluctuations, with notable peaks during periods of geopolitical tensions and economic disruptions.

Theme Two: State-wise and Commodity-wise Weightage in Calculating General Inflation Rate

Chart 2 : Distribution of Major Categories in Inflation Contribution

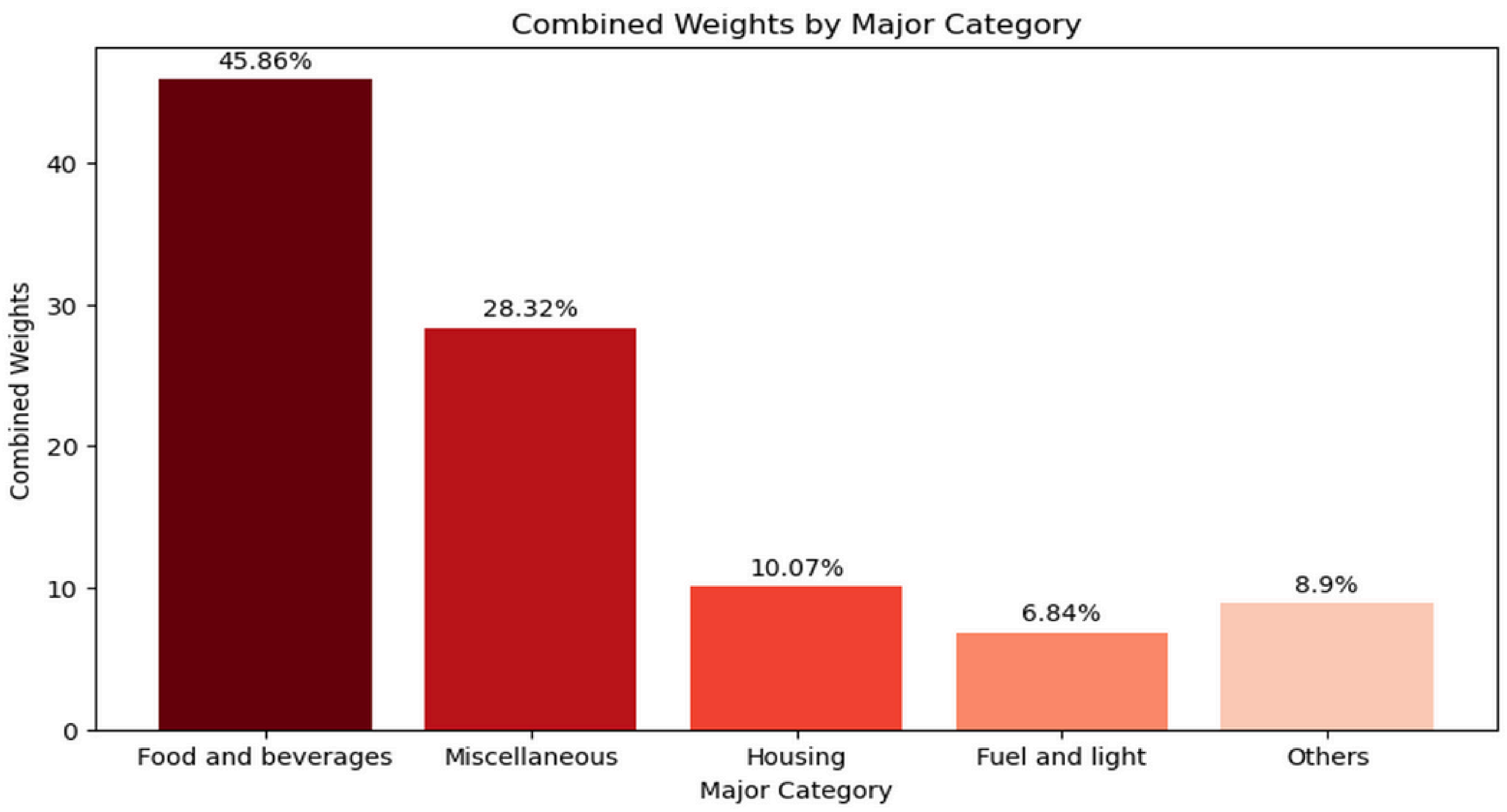


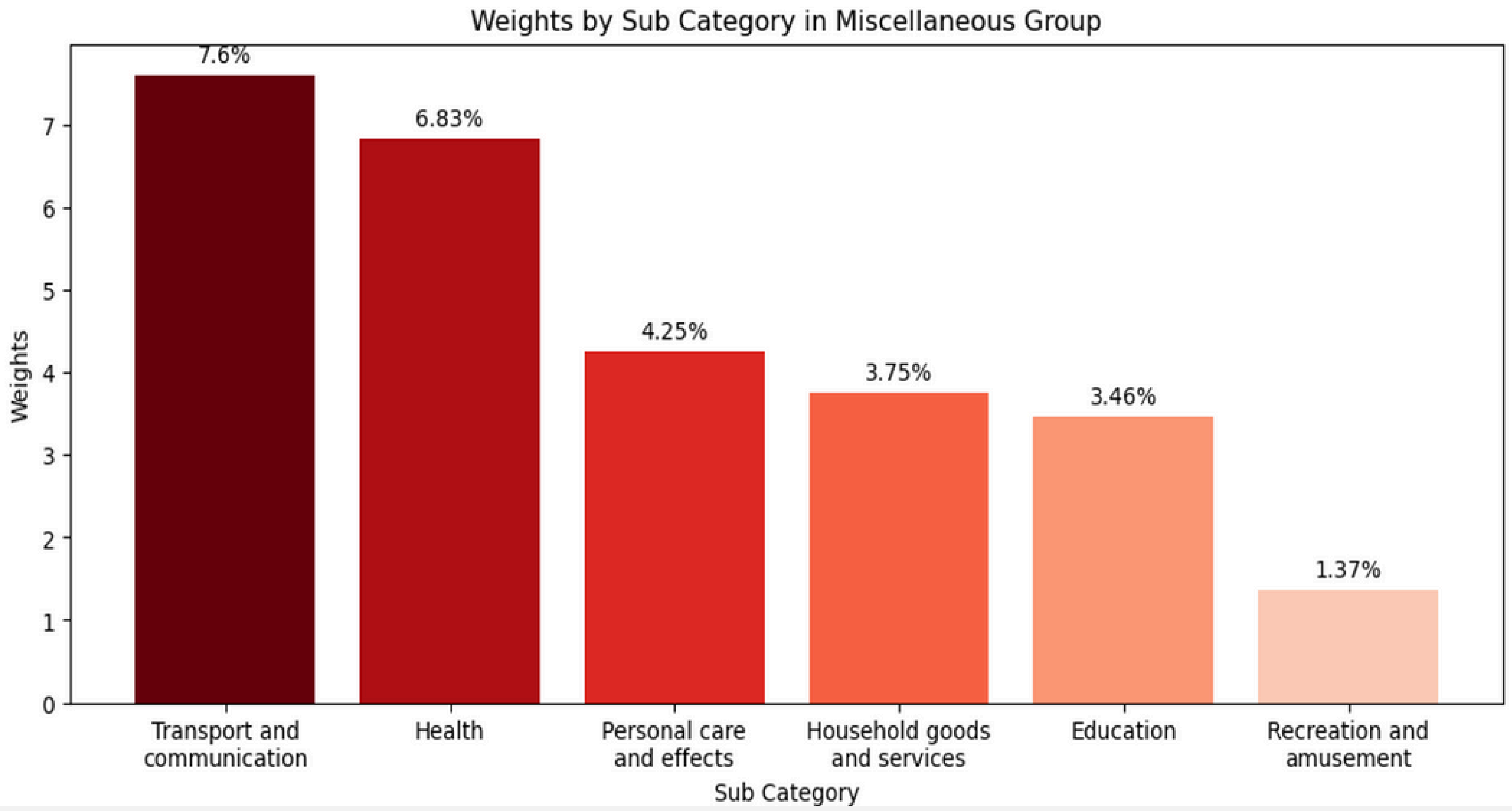
It shows the weightage assigned to each state for calculating India's general inflation rate. These weightages reflect economic significance and consumption patterns, based on factors like GDP contribution, population, and spending. States like Maharashtra (13.18%) and Uttar Pradesh (12.37%) have higher weightages, indicating their strong impact on national inflation.

Chart 3: Inflation trends across Indian states

State Name	2014	2015	2016	2017
	Year of Date			
Maharashtra	5.487	4.062	4.087	4.893
Uttar Pradesh	5.708	2.994	3.844	3.288
Tamil Nadu	4.859	3.529	3.564	4.265
West Bengal	6.318	3.254	4.250	3.421
Karnataka	5.926	5.775	4.837	3.629
Gujarat	5.657	4.235	4.862	3.225
Rajasthan	5.844	4.843	4.909	3.514
Bihar	6.462	3.830	3.229	3.176
Andhra Pradesh	5.547	4.547	5.110	3.673
Kerala	5.730	4.041	4.041	5.145

It highlights India's state-wise weightage distribution for inflation, where larger economies like Maharashtra and Uttar Pradesh have higher weightages due to their strong GDP and consumption, while smaller states like Sikkim and Mizoram have lower impact. These weightages are determined by NSO and RBI, considering factors like state GDP, household consumption, and demographics to ensure accurate inflation measurement

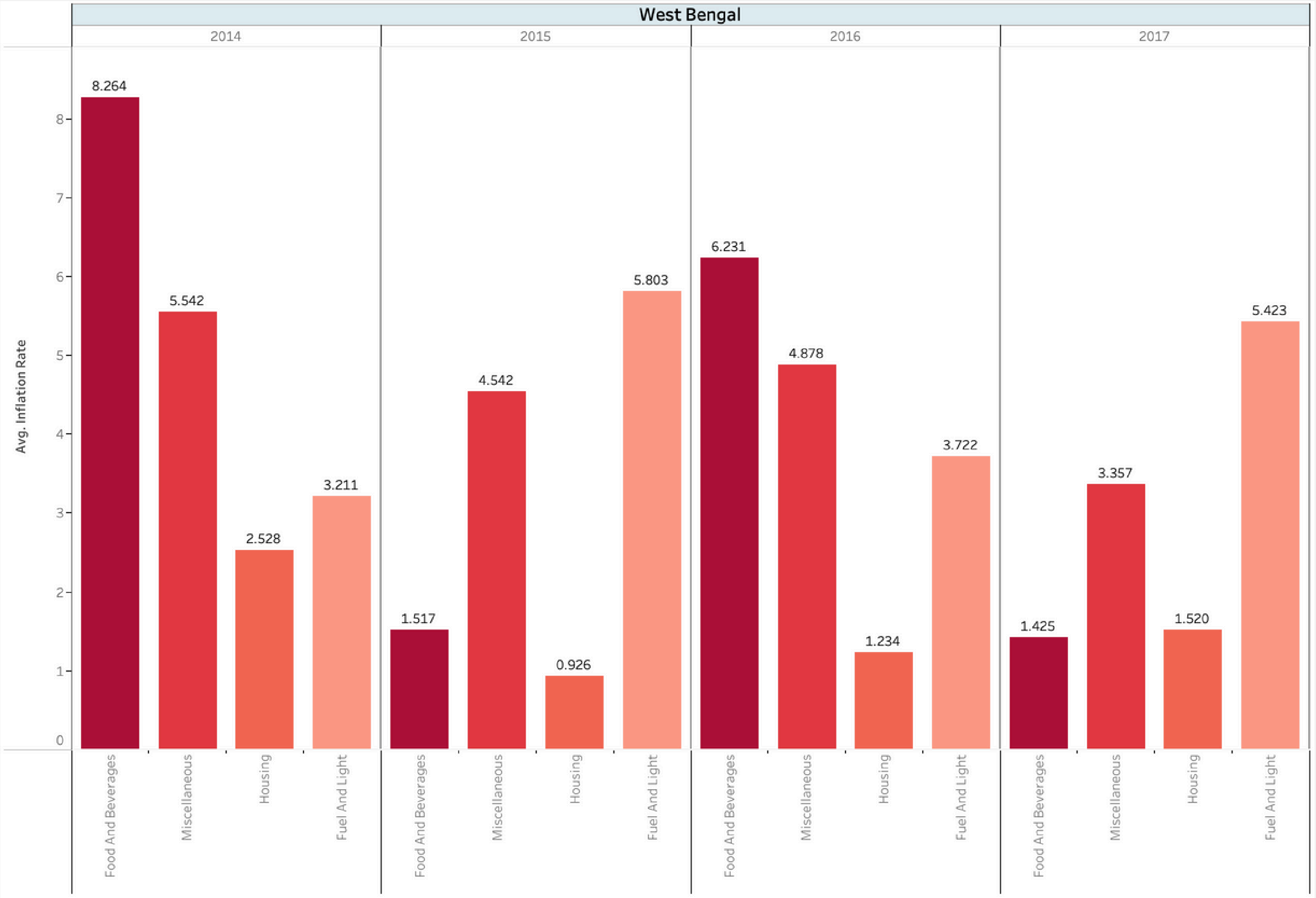




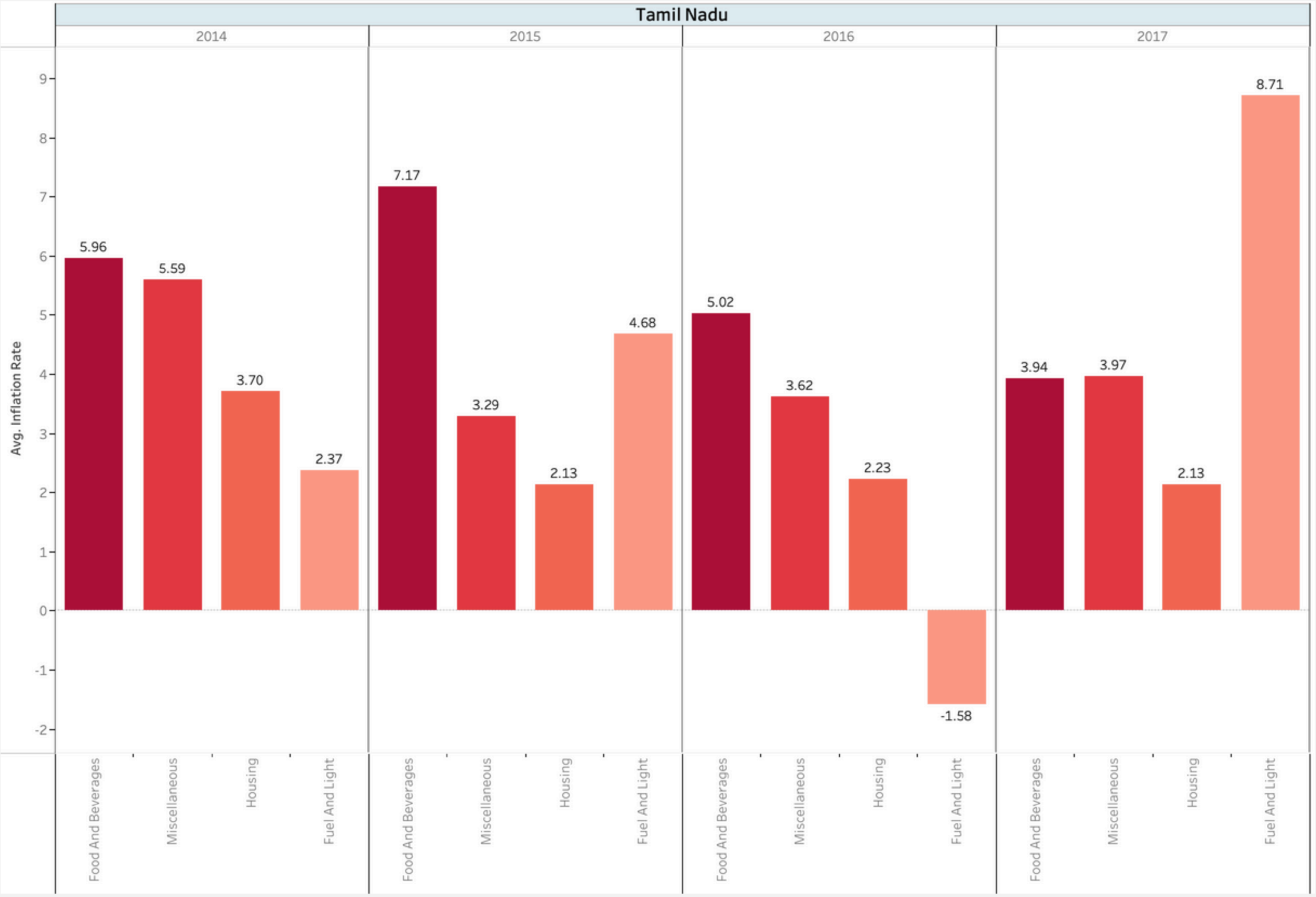
- Chart 4 shows the weightage distribution of commodities in inflation, with food and beverages (45.9%) having the highest impact, followed by housing (10.07%), and fuel & light (6.84%).
- The high weightage of food reflects its essential role in household expenses and sensitivity to factors like seasonal changes, supply disruptions, and agricultural productivity.
- Miscellaneous (28.32%) includes personal care, education, health, transport, and recreation, all of which influence inflation trends.
- The NSO assigns these weightages based on household consumption surveys to ensure inflation calculations accurately reflect living costs.

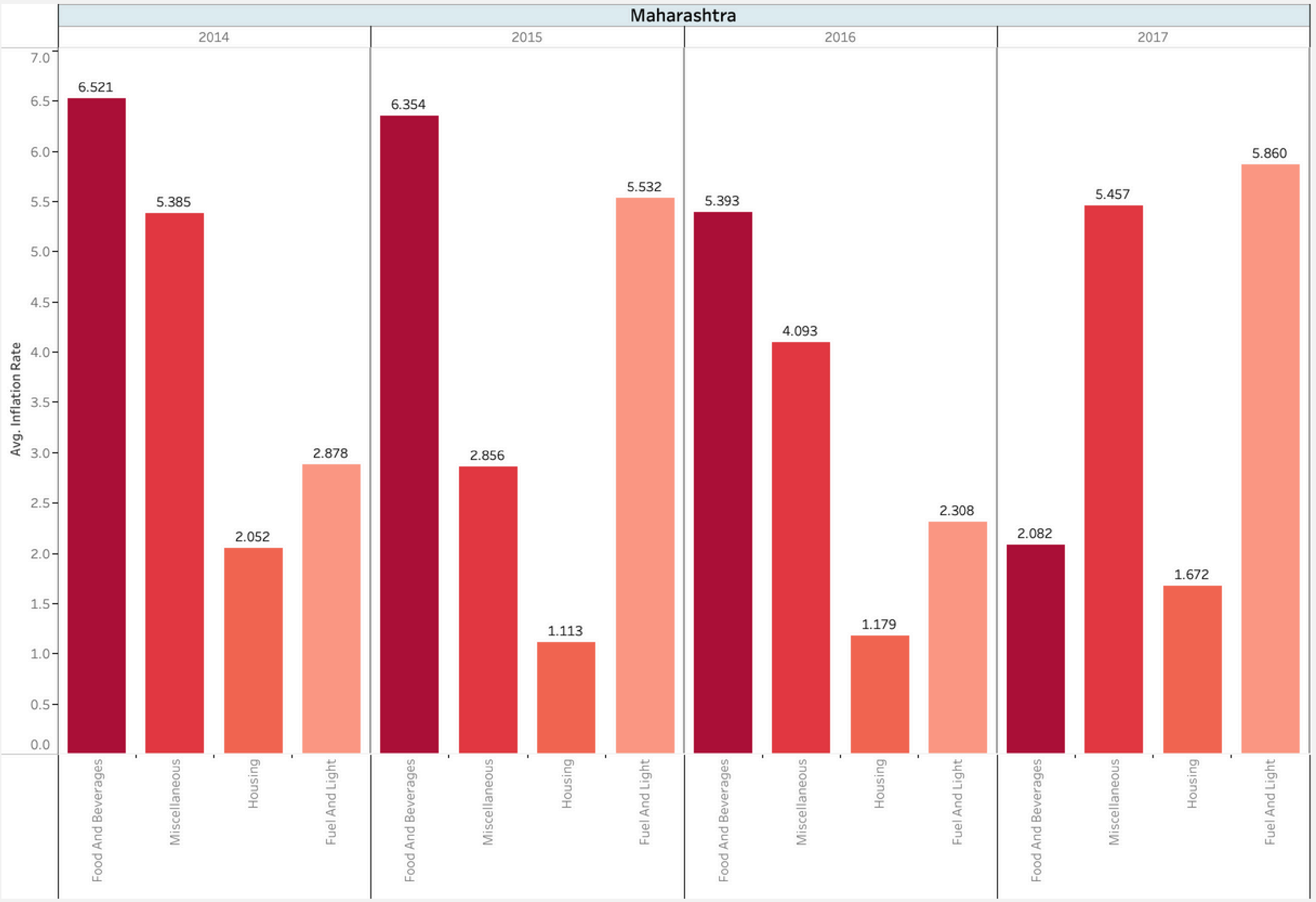
Theme Three: Analysing Inflation Dynamics in Key States (2014-2016)

Chart 5: Average Inflation Rate Across Key Sectors in Indian States (2014-2016)



- **Interrelationship with Housing:**
- **Interest Rates: Central banks often adjust interest rates to control overall inflation. Higher interest rates can make mortgages more expensive, thereby cooling housing demand and slowing housing inflation.**
- **Component of CPI: Housing costs are a significant component of the CPI, so high housing inflation can directly contribute to higher overall inflation.**
- **Wealth Effect: Rising home prices can increase homeowners' wealth, leading to increased consumer spending, which can drive overall inflation.**
- **Supply Chain: Issues in the supply chain, like higher costs for building materials, can lead to higher construction costs, which in turn affect housing prices and potentially overall inflation.**



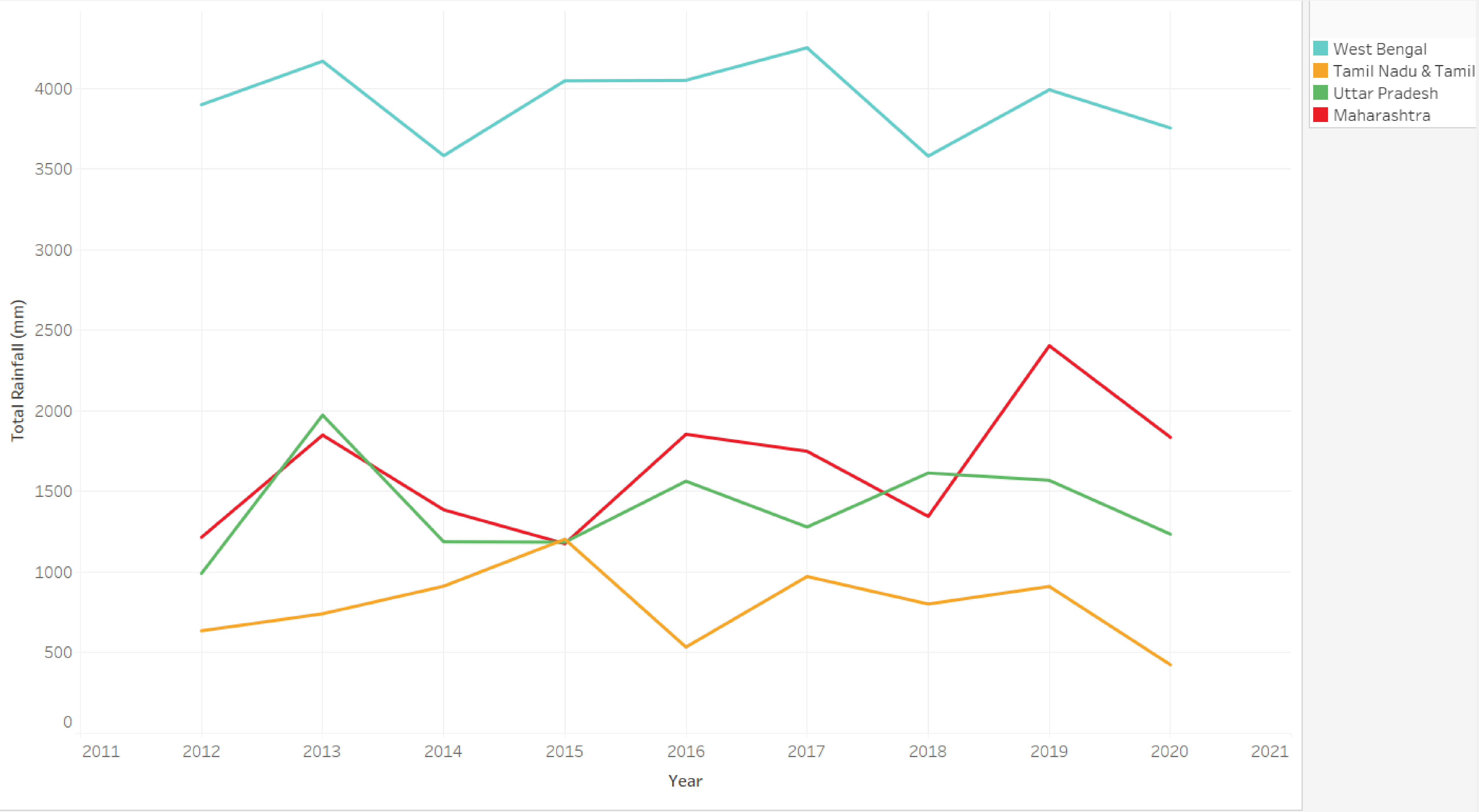


Interrelationship with the Fuel and Light sector:

- **Direct Impact: Fuel and energy costs are a significant component of the overall CPI. When fuel prices rise, they directly contribute to higher overall inflation.**
- **Supply Chain Costs: Higher fuel prices increase transportation and production costs across various industries. This can lead to higher prices for goods and services, thus driving overall inflation.**
- **Consumer Spending: When energy costs rise, consumers may have less disposable income to spend on other goods and services, potentially impacting overall demand and inflation.**
- **Geopolitical Factors: Geopolitical events, such as conflicts in oil-producing regions, can lead to spikes in fuel prices. These spikes can quickly translate into higher overall inflation due to the global nature of energy markets.**
- **Seasonal Variations: Seasonal demand, such as increased heating costs in winter or higher gasoline usage in summer, can cause fluctuations in fuel prices, impacting overall inflation rates**

Theme 4: Understanding Food and Beverage Inflation (Through Trade and Rainfall Trends in Key States)

Chart 6: Annual Rainfall Trends food and beverages in Selected Indian States (2012-2020)



- Food and Beverage Inflation is influenced by rainfall patterns and trade dynamics.
- Monsoon variations impact crop yields, directly affecting food prices.
- Import dependency on key food items can either stabilize or worsen inflation, depending on global price fluctuations.
- The analysis provides a comprehensive view of the key drivers of food inflation and its impact on overall inflation trends.

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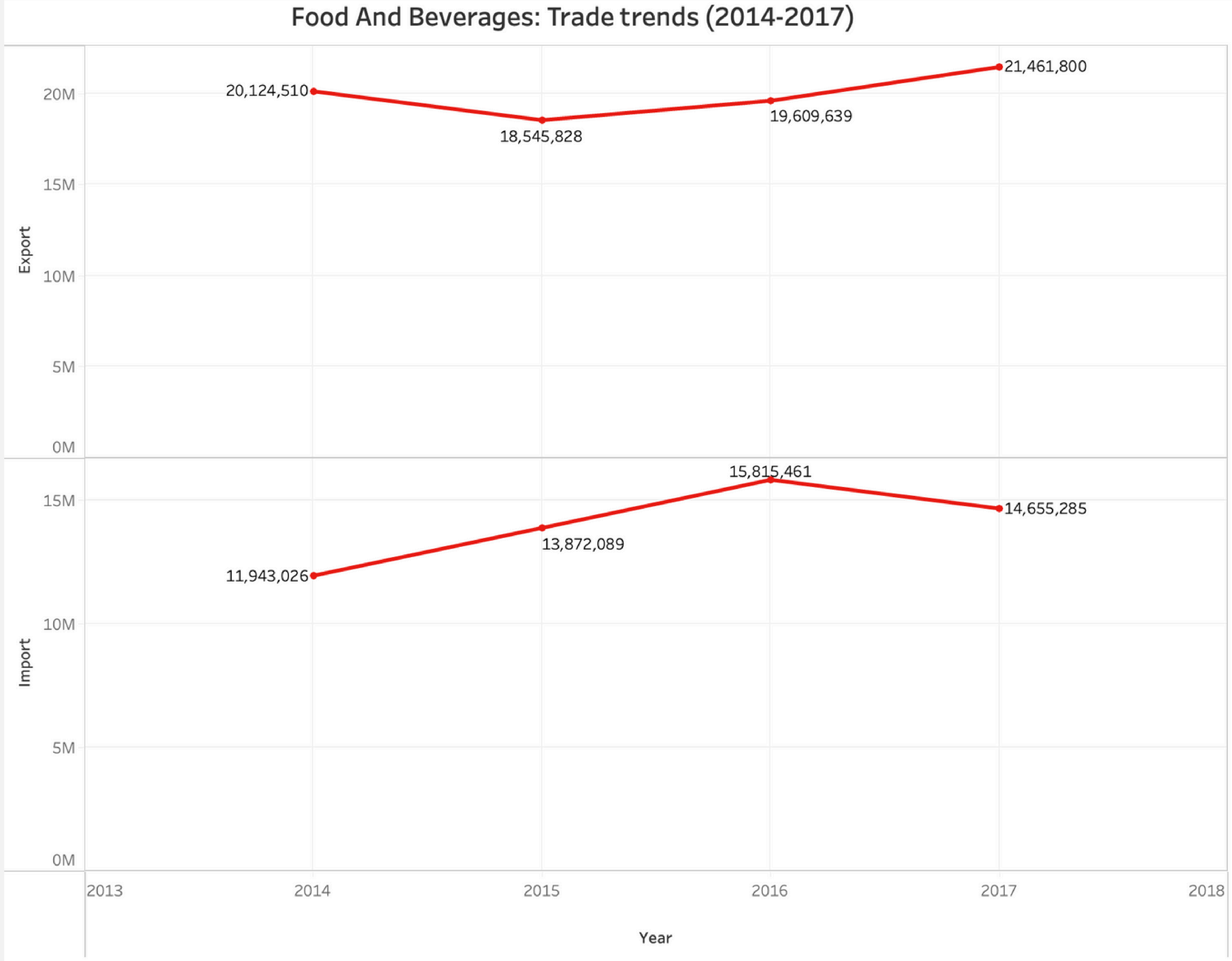
Chart 7: Trade Trends of food and beverages in Indian States (2014-2017)



Consistent Trade Surplus: From 2014 to 2017, the country maintained a positive trade balance in food and beverages, with exports consistently exceeding imports.

Export Trends: Exports dipped from $20.12 billion in 2014 to $18.54 billion in 2015 but rebounded, peaking at $21.46 billion in 2017, reflecting increased global demand and competitiveness.

Import Fluctuations: Imports were stable around $12 billion in 2014-2015, spiked to $15.82 billion in 2016—possibly due to higher domestic demand—and then declined to $14.66 billion in 2017.

Widening Trade Gap in 2017: The simultaneous rise in exports and drop in imports in 2017 led to the largest trade surplus in the period, highlighting strengthened global competitiveness.

Economic Implications: This positive trade balance boosts economic growth and is influenced by factors like global market dynamics, domestic policies, and currency fluctuations, indicating opportunities for further expansion.

KEY Takeaways

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1. **Inflation Was Sensitive to External Shocks:**

- **Global events like crude oil price fluctuations, the Russia-Ukraine war, and the COVID-19 pandemic significantly impacted inflation, particularly in fuel and food sectors.**

2. **Food & Beverage Inflation Was the Largest Contributor:**

- **Food and beverages, making up 45.9% of the CPI, were the main driver of inflation, with price volatility influenced by weather patterns, supply chains, and trade dependencies.**

3. **Rainfall Variability Affected Inflation:**

- **Deficient rainfall led to higher food prices, while good rainfall improved supply and lowered inflation, though extreme weather events caused unpredictability.**

4. **Trade Dependencies Increased Inflation Vulnerabilities:**

- **Heavy reliance on imports during deficient years raised food prices, but surplus production in favorable years reduced inflation. Trade policies lacked stability to buffer shocks.**

5. **State-Level Trends Were Diverse:**

- **States with higher agricultural dependency, like West Bengal and Tamil Nadu, experienced sharper inflation shifts, while some implemented effective policies to manage inflation.**

6. **Monetary & Fiscal Policies Stabilized Inflation:**

- **RBI's inflation-targeting helped control inflation, while fiscal policies like demonetization and GST caused short-term slowdowns. Government initiatives like crop insurance supported stability but varied in effectiveness.**

CONCLUSION

1. **Global & Domestic Factors Impacted Inflation:**

- **India's inflation trends were shaped by global shifts, domestic policies, climate variability, and trade dependencies.**

2. **RBI's Role in Stabilizing Inflation:**

- **RBI's inflation-targeting policy helped stabilize inflation, but challenges like import dependencies and fuel price volatility persisted.**

3. **State-Level Variations:**

- **States with strong agricultural productivity and trade resilience managed inflation better, while those dependent on rainfall faced higher volatility.**

4. **Future Focus for Inflation Control:**

- **Long-term solutions require investments in climate-resilient agriculture, efficient storage infrastructure, and AI-driven inflation forecasting.**

5. **Importance of Diversification & Modernization:**

- **Energy diversification and supply chain modernization are key to reducing the impact of external shocks on domestic prices.**

6. **Holistic Approach Needed:**

- **A coordinated approach integrating agriculture, trade, supply chain efficiency, and monetary policies is crucial for sustainable inflation control and economic resilience.**

Thank You